

1.
Which of the following best describes the C expression
`P++ -> s1.a` ?

- A. It increments the pointer P first, then accesses the member a of the structure s1 that the incremented pointer points to.
- B. It accesses the member a of the structure s1 that P currently points to, then increments the pointer P to the next structure in memory.
- C. It accesses the member of the next structure that P will point to after incrementing.
- D. It increments the member a of the structure s1 before accessing it

Answer: B

2.
Which of the following statements is true regarding slack bytes in a structure?

- A. Slack bytes are used to store hidden information about the structure.
- B. Slack bytes can cause unexpected behaviour when copying the structure using `memcpy`.
- C. Slack bytes can be safely ignored since they don't affect data access within the structure.
- D. Slack bytes help ensure that members are properly aligned for faster access on specific hardware.

Answer: D

3. What will be the output of the following C code?

```
#include<stdio.h>
int main(void)
{
    struct sunbeam
    {
        struct
        {
            int a1;
        }s1;
        struct
        {
            int a2;
        }s2;
    }a[2] ,*p;

    a[1].s1.a1=10;
    p = a;
    p->s2.a2= p->s1.a1;

    printf("%d %d\n",p[-1].s2.a2 , p[1].s1.a1);

    return 0;
}
```

- A. Garbage value , 10
- B. Garbage value , Garbage value
- C. Compile Time Error
- D. 10 , 10

Answer: A

4. What will be the output of the following C code?

```
#include<stdio.h>
union demo
{
    int a1;
    unsigned k1:2;
    unsigned char s1:6;
}st;

int main(void)
{
    st.s1=127;
    st.k1=8;
    st.a1 = 15;
    int *p = st.a1;

    ++*p;

    printf("%d %d %d ",st.a1,st.k1,st.s1);
    printf("Size : %u %u ",sizeof(st.a1),sizeof(st.k1));

    return 0;
}
```

- A. 63 , 0 , 8 , 4 , 4
- B. 15 , 0 , 15 , 4 , 4
- C. 16, 0 , 16 , 4 , 4
- D. Error

Answer: D

5. What will be the output of the following C code?

```
#include <stdio.h>
typedef struct demo
{
    int a1;
    char ch;
}demo_t;

int main()
{
    demo_t s1 = {1};
    demo_t s2 = {1};

    if(s1 == s2)
        printf("sunbeam Karad\n");
    else
        printf("Sunbeam Pune MarketYard\n");

    return 0;
}
```

- A.sunbeam Karad
- B.Sunbeam Pune MarketYard
- C.Compile time Error
- D.RunTime Error

Answer : C